PRACTICAL –10

Aim: Implement Kruskal's Algorithm.

Algorithm:

Algorithm:
Input: No. of vertices, No. of edges
Output: Edges in minimum spanning tree & min cost.

⇒ Start
⇒ Read the no. of vertices & edges.
⇒ Store all edges with their weights
⇒ Sort the edges in ascending order of weight.
⇒ Select the edge with the min weight.
⇒ check whether adding the edge form a cycle.
⇒ If it does not form a cycle add the edge to the minimum spanning tree.
⇒ Repeat steps 5-7 until V-1 edges are selected where V is the no. of vertices.
⇒ Display the selected edges and the total min cost
⇒ Stop.

Progarm:

```
#include <iostream>

#include <algorithm>

using namespace std;


struct Edge {
```

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```
    int u, v, weight;
};

bool compare(Edge a, Edge b) {
    return a.weight < b.weight;
}

int findParent(int parent[], int i) {
    if (parent[i] == i)
        return i;
    return findParent(parent, parent[i]);
}

int main() {
    int n, e;
    Edge edges[100];

    cout << "Enter number of vertices: ";
    cin >> n;

    cout << "Enter number of edges: ";
    cin >> e;
```

```
cout << "Enter edges (source destination weight):\n";

for (int i = 0; i < e; i++) {
    cin >> edges[i].u >> edges[i].v
        >> edges[i].weight;
}

sort(edges, edges + e, compare);

int parent[100];

for (int i = 1; i <= n; i++) {
    parent[i] = i;
}

int minCost = 0;
int count = 0;

cout << "\nEdges in Minimum Spanning Tree:\n";

for (int i = 0; i < e && count < n - 1; i++) {
```

```
        int u = findParent(parent, edges[i].u);
        int v = findParent(parent, edges[i].v);

        if (u != v) {
            cout << edges[i].u << " - "
                 << edges[i].v << " : "
                 << edges[i].weight << endl;

            minCost += edges[i].weight;
            parent[u] = v;
            count++;
        }
    }

    if (count != n - 1) {
        cout << "Graph is disconnected.\n";
    } else {
        cout << "Minimum Cost = " << minCost << endl;
    }

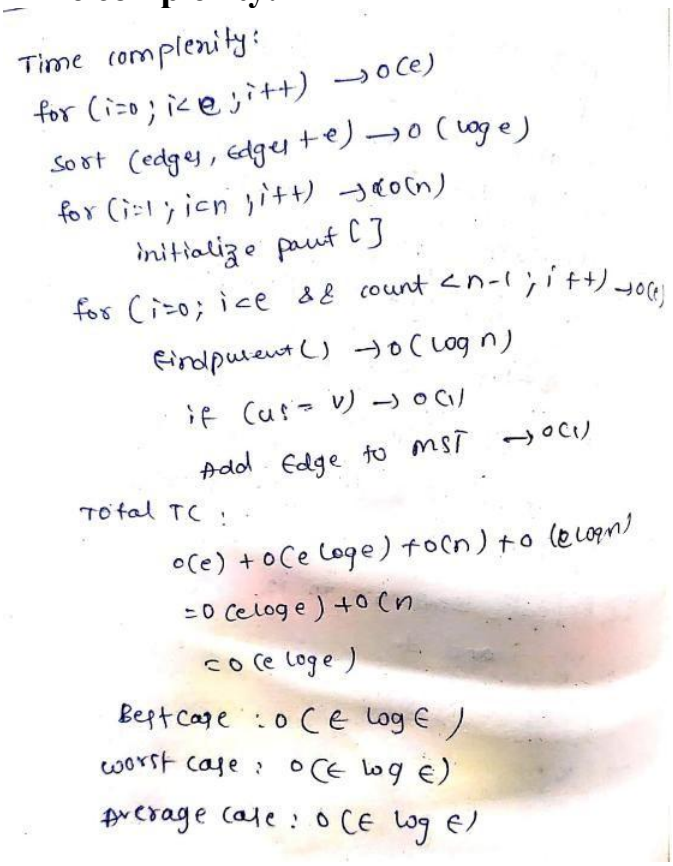
    return 0;
}
```

Output:

```
Enter number of vertices: 4
Enter the adjacency matrix:
0 2 0 6
2 0 3 8
0 3 0 5
6 8 5 0

Edges in Minimum Spanning Tree:
1 - 2 : 2
2 - 3 : 3
3 - 4 : 5
Minimum Cost = 10
```

Time complexity:



Conclusion: Kruskal's algorithm has a time complexity of (O(Elog E), mainly because it sorts all edges in ascending order of their weights.

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